



UNIVERSITY OF THE PUNJAB

BS (After Associate Degree) Sixth Semester – Fall 2023

Paper: Mathematical Methods of Physics-II

(Common with BS4Y 6th)

Course Code: PHYS-3503

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Solve the following:

(6x5=30)

- 1) By using the Levi-Civita symbol, evaluate the determinant of the given matrix

$$\begin{pmatrix} 2 & 1 & -3 \\ 3 & 4 & 0 \\ 1 & -2 & 1 \end{pmatrix}$$

- 3 2) Any arbitrary tensor of rank two can be expressed as a sum of symmetric and anti-symmetric tensor of rank two. 1

- 5 3) Show that every cyclic group is abelian 5

- 5 4) Show that the set $S = \{1, 3, 5, 7\}$ is a group under multiplication (mod 8). 3

- 5 5) Expand $f(z) = \frac{1}{z(z-1)}$ in a Laurent series valid for the annular domain $1 < |z|$.

- 6) By using Cauchy's residue theorem to evaluate the given integral along the indicated contour

$$\oint_C \frac{\tan z}{z} dz$$

where $C: |z - i| = 2$.

Q.2. Solve the following.

(5x6=30)

- 1) Evaluate the Cauchy principal value of

$$\int_{-\infty}^{\infty} \frac{1}{x^4 + 1} dx$$

- 2) By using Residue theorem, evaluate the integral

$$\int_0^{2\pi} \frac{d\theta}{1 + a \cos \theta}$$

where $|a| < 1$.

- 4 3) Show that the T_{ij} given by $T = [T_{ij}] = \begin{pmatrix} x_2^2 & -x_1 x_2 \\ -x_1 x_2 & x_1^2 \end{pmatrix}$ are the components of a second order tensor. 3

- 4) Show that the quantities $g_{ij} = e_i \cdot e_j$ form the covariant components of a second-order tensor.

- 5 5) If every element of a group G is its own inverse, show that G is abelian. 6



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answer of the following questions: (6x5=30)

1. What is Kronecker delta? Prove that $\epsilon_{ijk} \epsilon_{ijk} = 6$.
2. Apply tensor to show that $\vec{a} \cdot \vec{b} \times \vec{c} = \vec{b} \cdot \vec{c} \times \vec{a} - \vec{c} \cdot \vec{a} \times \vec{b}$.
3. Prove that a finite group of prime order is always cyclic.
4. Show that an infinite cyclic group has exactly two different generators.
5. Evaluate the integral $\oint_C \frac{e^{2z} + \sin z}{(z+1)^4} dz$ where C encloses the point $z = -1$.
6. A function $f(z)$ which has no singularity in the finite part of the plane or at infinity is a constant.

Q.2. Solve the following.

(5x6=30)

1. Show that the transformation from X to X' frame is

$$X'_1 = \frac{1}{3}(2X_1 + 2X_2 - X_3), \quad X'_2 = \frac{1}{3}(2X_1 - X_2 + 2X_3)$$

$$X'_3 = \frac{1}{3}(-X_1 + 2X_2 + 2X_3)$$

is orthogonal. Also prove that $A'_{ii} = A_{pp}$.

2. State and prove Lagrange's theorem. Also define Christoffel symbols.

3. Show that $\text{Aut}(Z_n)$ is isomorphic to $U(n)$ where $n \in \mathbb{Z}^+$.

4. Evaluate $\int_0^\pi \frac{1+2\cos\theta}{5+4\cos\theta} d\theta$.

5. State and prove Cauchy's Residue Theorem.



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Solve the following:

(6x5=30)

1. What is dummy index? Expand $a_{pqr} X_p X_{qq}$.
2. Show that the product of a tensor and a Pseudo tensor is also a Pseudo tensor.
3. State and prove Lagrange theorem.
4. Prove that every cyclic group is Abelian.

5. Prove that the function
$$f(Z) = \frac{x^3(1+i) - y^3(1-i)}{x^2 + y^2} \quad \text{when } Z \neq 0$$

$$= 0 \quad \text{when } Z = 0$$

6. Expand $\log(1+Z)$ in a Taylor's series about $Z=0$ and determine the region of convergence for the resulting series.

Q.2. Solve the following.

(5x6=30)

1. The transformation from X to X' frame is

$$X'_1 = -\frac{3}{7}X_1 - \frac{6}{7}X_2 - \frac{2}{7}X_3, \quad X'_2 = -\frac{2}{7}X_1 + \frac{3}{7}X_2 - \frac{6}{7}X_3,$$

$$X'_3 = \frac{6}{7}X_1 - \frac{2}{7}X_2 - \frac{3}{7}X_3.$$

Find the components of 3rd rank tensor

where all components (except $A_{111} = A_{222} = 1$ and $A_{212} = -2$) are zero.

2. Show that $A_{pq} = \begin{pmatrix} x_2^2 & x_1x_2 \\ -x_1x_2 & x_1^2 \end{pmatrix}$ are not the components of a 2nd order tensor.

3. Prove that the set of all automorphisms of a group G is a group under the operation of composition of functions

4. Find the Residues of $f(Z) = \frac{e^Z}{Z^2(Z - \pi i)^4}$.

5. Evaluate $\int_0^\infty \frac{\cos mx}{a^2 + x^2} dx$ and deduce the value of $\int_0^\infty \frac{x \sin mx}{a^2 + x^2} dx$ where a and m are constants.

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answer of the following questions:

(6x5=30)

1. What is dummy index? Expand $a_{pqr} X_p X_{qq}$.
2. Show that the product of a tensor and a Pseudo tensor is also a Pseudo tensor.
3. State and prove Lagrange theorem.
4. Prove that every cyclic group is Abelian.

$$f(Z) = \frac{x^3(1+i) - y^3(1-i)}{x^2 + y^2} \quad \text{when } Z \neq 0$$

5. Prove that the function

$$= 0 \quad \text{when } Z = 0$$

6. Expand $\log(1+Z)$ in a Taylor's series about $Z=0$ and determine the region of convergence for the resulting series.

Q.2. Solve the following.

(5x6=30)

1. The transformation from X to X' frame is

$$X'_1 = -\frac{3}{7}X_1 - \frac{6}{7}X_2 - \frac{2}{7}X_3, \quad X'_2 = -\frac{2}{7}X_1 + \frac{3}{7}X_2 - \frac{6}{7}X_3,$$

$$X'_3 = \frac{6}{7}X_1 - \frac{2}{7}X_2 - \frac{3}{7}X_3. \quad \text{Find the components of } 3^{\text{rd}} \text{ rank tensor}$$

where all components (except $A_{111} = A_{222} = 1$ and $A_{212} = -2$) are zero.

2. Show that $A_{pq} = \begin{pmatrix} x_2^2 & x_1x_2 \\ -x_1x_2 & x_1^2 \end{pmatrix}$ are not the components of a 2nd order tensor.

3. Prove that the set of all automorphisms of a group G is a group under the operation of composition of functions

4. Find the residues of $f(z) = \frac{e^z}{z^2(z-\pi i)^4}$.

5. Evaluate $\int_0^{\infty} \frac{\cos \frac{nx}{a^2+x^2}}{x^2} dx$ and deduce the value of $\int_0^{\infty} \frac{x \sin mx}{a^2+x^2} dx$ where a and m are constants.



Q.1. Solve the following:

(6x5=30)

1. With $\hat{e}_1$ a unit vector in the direction of increasing u_1 , show that

$$\nabla \cdot \hat{e}_1 = \frac{1}{h_1 h_2 h_3} \frac{\partial}{\partial u_1} (h_2 h_3)$$

2. Show that $\mathbf{r} = \rho \hat{e}_\rho + z \hat{e}_z$. Working entirely in circular cylindrical coordinates also prove that $\nabla \cdot \mathbf{r} = 3$ and $\nabla \times \mathbf{r} = 0$. (Note: $x = \rho \cos \phi$, $y = \rho \sin \phi$, $z = z$)

3. Given $A_k = \frac{1}{2} \epsilon_{ijk} B^{ij}$, with $B^{ij} = -B^{ji}$, antisymmetric tensor, show that

$$B^{mn} = \epsilon^{mnk} A_k.$$

4. Show that

$$\frac{1}{2\pi i} \oint_C z^{m-n-1} dz, \quad m \text{ and } n \text{ integers}$$

(with the contour encircling the origin once counterclockwise) is a representation of the Kronecker δ_{mn} .

5. Evaluate

$$\oint_{|z-2i|=4} \frac{z}{z^2+9} dz,$$

by using Cauchy's integral formula.

6. Use the divergence theorem to establish Green's identity

$$\iint_S (f \nabla g) \cdot \mathbf{n} dS = \iiint_D (f \nabla^2 g + \nabla f \cdot \nabla g) dV,$$

here f and g are scalar functions with continuous second partial derivatives.

Q.2. Solve the following.

(5x6=30)

1. Evaluate the Cauchy principal value of

$$\int_{-\infty}^{+\infty} \frac{dx}{(x^2 + a^2)^2} (a > 0).$$

2. The derivative of the second-order tensor $\mathbf{T}$ with respect to the coordinate u^k , find an expression for the covariant derivative $T^i_j{}_{;k}$ of its contravariant components.

3. Calculate the elements g_{ij} of the metric tensor for spherical polar coordinate. Also calculate the Γ^k_{ij} for spherical polar coordinates. (Note: $x = r \cos \phi \sin \theta$, $y = \rho \sin \phi \sin \theta$, $z = r \cos \theta$)

4. Use the Cauchy's residue theorem to evaluate

$$\oint_{|z-1|=2} \frac{\tan z}{z} dz.$$

5. Find the spherical coordinate components of the velocity and acceleration of a moving particle. ($x = r \cos \phi \sin \theta$, $y = \rho \sin \phi \sin \theta$, $z = r \cos \theta$)



ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Solve the following:

(5×4=20)

- ✓1. Show that an arbitrary tensor (neither symmetric nor antisymmetric) can always be written as sum of a symmetric tensor and an antisymmetric tensor.
- ✓2. Evaluate the surface integral $I = \int_S \mathbf{a} \cdot d\mathbf{S}$, where $\mathbf{a} = x\mathbf{i}$ and S is the surface of the hemisphere $x^2 + y^2 + z^2 = a^2$ with $z \geq 0$.
- ✓3. Show that the quantities $g_{ij} = \mathbf{e}_i \cdot \mathbf{e}_j$ form the covariant components of a second-rank tensor.
- ✓4. Determine the order of the poles of $f(z) = \frac{\sin z}{z^2 - z}$.
- ✓5. Expand $f(z) = \frac{1}{z(z-1)}$ in a Laurent series valid for $1 < |z|$.

Q.3. Solve the following.

(5×6=30)

- ✓1. Evaluate the Cauchy principal value of

$$\int_{-\infty}^{+\infty} \frac{x \sin x}{x(x^2 + 2)} dx.$$

- ✓2. Prove the following identities

$$(\mathbf{e}^i \cdot \mathbf{e}^j)(\mathbf{e}_j \cdot \mathbf{e}_k) = \delta_k^i.$$

$$\Gamma_{jk}^m = \Gamma_{kj}^m.$$

- ✓3. Calculate the basis vectors $\mathbf{e}_r, \mathbf{e}_\theta$, and $\mathbf{e}_\phi$. Also compute the basis vectors $\mathbf{e}_r, \mathbf{e}_\theta$ and $\mathbf{e}_\phi$. Further Show that $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_\phi\}$ and $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_\phi\}$ are reciprocal systems of vectors. ($x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$ and $z = r \cos \theta$).

- ✓4. Use an indented contour and residues, show that

$$\text{P.V.} \left(\int_{-\infty}^{+\infty} \frac{\sin x}{x} dx \right) = \pi.$$

- ✓5. Suppose that C is a piecewise-smooth simple closed curve bounding a simple connected region R . If the functions $P(x, y)$ and $Q(x, y)$ and their partial derivatives are continuous on R , then show that

$$\oint_C (P(x, y)dx + Q(x, y)dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$



UNIVERSITY OF THE PUNJAB

Fifth Semester – 2019

Examination: B.S. 4 Years Program

Roll No. 019599.....

PAPER: Mathematical Methods of Physics-

Course Code: PHY-302 Part – II

MAX. TIME: 2 Hrs. 45 Min.

MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Questions with Short Answers.

(5x4=20)

1. If $\hat{e}_1$ a unit vector in the direction of increasing u_1 , show that

$$\nabla \cdot \hat{e}_1 = \frac{1}{h_1 h_2 h_3} \frac{\partial(h_2 h_3)}{\partial u_1}$$

2. Use the divergence theorem to establish following identities

$$\iint_S (f \nabla g) \cdot n dS = \iiint_D (f \nabla^2 g + \nabla f \cdot \nabla g) dV$$

here f and g are scalar functions with continuous second partial derivatives..

3. Given $A_k = \frac{1}{2} \epsilon_{ijk} B^{ij}$, with $B^{ij} = -B^{ji}$, antisymmetric tensor, show that

$$B^{mn} = \epsilon^{mnk} A_k$$

4. Determine the order of the poles of $f(z) = \frac{\cot(\pi z)}{z^2}$.

5. Suppose z_0 is any constant complex number interior to any simple closed contour C . Show that

$$\oint_C \frac{dz}{(z - z_0)^n} = \begin{cases} 2\pi i, & n = 1 \\ 0, & n \text{ is a positive integer } \neq 1 \end{cases}$$

Q.3. Questions with Brief Answers.

(5x6=30)

1. Show that

$$\int_0^\pi \frac{d\theta}{(a + b \cos \theta)^2} = \frac{\pi a}{(a^2 - 1)^{3/2}}$$

2. Show that

$$\Gamma_{ij}^m = \frac{1}{2} g^{mk} \left(\frac{\partial g_{jk}}{\partial u^i} + \frac{\partial g_{ki}}{\partial u^j} - \frac{\partial g_{ij}}{\partial u^k} \right)$$

3. Using Cauchy's integral formula for derivatives, evaluate

$$\oint \frac{z+1}{z^4 + 4z^3} dz,$$

where C is the circle $|z| = 1$.

4. Evaluate

$$\int_{-\infty}^{+\infty} \frac{x^2}{1+x^4} dx.$$

5. Find the spherical coordinate components of the velocity and acceleration of a moving particle. ($x = r \cos \phi \sin \theta$, $y = \rho \sin \phi \sin \theta$, $z = r \cos \theta$)



UNIVERSITY OF THE PUNJAB

Fifth Semester 2018
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Mathematical Methods of Physics-
Course Code: PHY-302

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.
SUBJECTIVE TYPE

Section-II (Short Questions)

Marks=20

1. Show that $\{e_i\}$ and $\{\epsilon_i\}$ are reciprocal systems of vectors.
2. Expand $f(z) = e^{3/z}$ in a Laurent series valid for $0 < |z|$.
3. Show that $z = 0$ is an essential singularity of $f(z) = z^3 \sin(1/z)$.
4. Show that $\mathbf{r} = \rho \hat{\mathbf{e}}_\rho + z \hat{\mathbf{e}}_z$, also prove that $\nabla \cdot \mathbf{r} = 3$ and $\nabla \times \mathbf{r} = 0$. (Note: $x = \rho \cos \phi$, $y = \rho \sin \phi$, $z = z$)
5. Using the calculus of residues, show that $\int_0^\pi \cos^{2n} \theta d\theta = \pi \frac{(2n)!}{2^{2n} (n!)^2}$, $n = 0, 1, 2, \dots$

Section-III

Marks=30

1. Evaluate

$$\int_0^{2\pi} \frac{\cos(3\theta) d\theta}{5 - 4 \cos \theta}.$$

2. Evaluate

$$\oint_{|z-2i|=4} \frac{z}{z^2+9} dz,$$

by using Cauchy's integral formula.

3. By considering the derivative of the second-order tensor $\mathbf{T}$ with respect to the coordinate u^k , find an expression for the covariant derivative $T_{ij,k}$ of its contravariant components.
4. The electric field $\mathbf{E} = -\nabla\varphi$; this is derived from a scalar, the electrostatic potential φ , and has components $E_i = -\frac{\partial\varphi}{\partial x_i}$. Show that $\mathbf{E}$ is a first order tensor.
5. Find the circular cylindrical components of the velocity and acceleration of a moving particle. (Hint: $\mathbf{r}(t) = \rho(t)\hat{\mathbf{e}}_\rho(t) + z(t)\hat{\mathbf{e}}_z$ and $x = \rho \cos \phi$, $y = \rho \sin \phi$, $z = z$)



UNIVERSITY OF THE PUNJAB

Fifth Semester 2017
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Mathematical Methods of Physics-
Course Code: PHY-302

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Section-II (Short Questions)

Marks=20

1. If $\mathbf{B} = \nabla \times \mathbf{A}$, show that

$$\oint \mathbf{B} \cdot d\boldsymbol{\sigma} = 0,$$

for any closed surface S .

2. Expand

$$f(z) = \frac{1}{z(1-z)^2}$$

in a Laurent series valid for $0 < |z| < 1$.

3. Determine whether $z = 0$ is an isolated or non-isolated singularity of $f(z) = \tan(1/z)$.
4. A vector $\mathbf{A}$ is decomposed into a radial vector $\mathbf{A}_r$ and a tangential vector $\mathbf{A}_t$. If $\hat{\mathbf{r}}$ is a unit vector in the radial direction, show that

$$\begin{aligned} \mathbf{A}_r &= \hat{\mathbf{r}}(\mathbf{A} \cdot \hat{\mathbf{r}}), \\ \mathbf{A}_t &= -\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \mathbf{A}). \end{aligned}$$

5. Evaluate $\oint_C \frac{z+1}{z^2+4z} dz$, where C is the circle $|z| = 1$.

Section-III

Marks=30

1. Show that

$$\int_0^\pi \frac{d\theta}{(a + \cos \theta)^2} = \frac{a\pi}{(\sqrt{a^2 - 1})^3}, \quad (a > 1).$$

2. Use Cauchy's residue theorem to evaluate

$$\oint_{|z|=\frac{1}{2}} \frac{\cot \pi z}{z^2} dz.$$

3. Suppose $A = [a_{ij}]$, $B = [b_{ij}]$ and that $B = A^{-1}$. By considering the determinant $a = |A|$, show that

$$\frac{\partial a}{\partial u^k} = ab^{jk} \frac{\partial a_{ij}}{\partial u^k}.$$

4. We may define Christoffel symbols of the first kind by

$$\Gamma_{ijk} = g_{il} \Gamma_{jk}^l.$$

Show that these are given by

$$\Gamma_{kij} = \frac{1}{2} \left\{ \frac{\partial g_{ik}}{\partial u^j} + \frac{\partial g_{jk}}{\partial u^i} - \frac{\partial g_{ij}}{\partial u^k} \right\}.$$

5. Prove the expression $\nabla \cdot \mathbf{a} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} (h_2 h_3 a_1) + \frac{\partial}{\partial u_2} (h_1 h_3 a_2) + \frac{\partial}{\partial u_3} (h_1 h_2 a_3) \right]$ in orthogonal curvilinear coordinates.



UNIVERSITY OF THE PUNJAB

Fifth Semester 2016
Examination: B.S. 4 Years Programme

Roll No. 029583

PAPER: Mathematical Methods of Physics-
Course Code: PHY-302

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.
SUBJECTIVE TYPE

Section-II (Short Questions)

Marks=20

1. Use the divergence theorem to establish Green's identity

$$\iint_S (f \nabla g) \cdot n dS = \iiint_D (f \nabla^2 g + \nabla f \cdot \nabla g) dV,$$

where f and g are scalar functions with continuous second partial derivatives.

2. Expand

$$f(z) = \frac{z}{(z+1)(z-2)}$$

in a Laurent series valid for $0 < |z-2| < 3$.

3. With $\hat{e}_1$ a unit vector in the direction of increasing u_1 , show that

$$\nabla \cdot \hat{e}_1 = \frac{1}{h_1 h_2 h_3} \frac{\partial}{\partial u_1} (h_2 h_3)$$

4. If A is irrotational (i.e. $\nabla \times A = 0$), show that $A \times r$ is solenoidal (i.e. $\nabla \cdot (A \times r) = 0$).

5. Show whether or not the function $f(z) = \text{Re}(z)$ is analytic.

Section-III

Marks=30

1. Show that

$$\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} = \frac{2\pi}{\sqrt{a^2 - b^2}}, \quad (a > b > 0).$$

2. Use the Cauchy's residue theorem to evaluate

$$\oint_{|z-1|=2} \frac{\tan z}{z} dz.$$

3. Find the circular cylindrical polar components of the velocity and acceleration of a moving particle. ($x = \rho \cos \phi$, $y = \rho \sin \phi$, $z = z$)

4. Calculate the elements g_{ij} of the metric tensor for spherical polar coordinate. ($x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$)

5. By considering the derivative of the second-order tensor T with respect to the coordinate u^k , find an expression for the covariant derivative $T^i_j{}_{;k}$ of its contravariant components.



UNIVERSITY OF THE PUNJAB

Fifth Semester 2015
Examination: B.S. 4 Years Programme

Roll No. 9922

PAPER: Mathematical Methods of Physics-
Course Code: PHY-302

TIME ALLOWED: 2 hrs. & 30 mi
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.
SUBJECTIVE TYPE

Section-II (Short Questions)

Marks=20

1. The double summation $K_{ij}A^iB^j$ is invariant for any two vectors A^i and B^j . Prove that K_{ij} is a second rank tensor.
2. Expand

$$f(z) = \frac{1}{(z-1)^2(z-3)}$$

in a Laurent series valid for $0 < |z-3| < 2$.

3. If a vector function depends on both space coordinates (x, y, z) and time t , (that is $F = F(x, y, z, t)$), show that

$$dF = (dr \cdot \nabla)F + \frac{\partial F}{\partial t} dt$$

Show that the covariant derivative of the metric tensor is zero (i.e. $g_{ij,k} = g_{ik,j} = 0$).

5. If there is some common region in which $w_1 = u(x, y) + iv(x, y)$ and $w_2 = w_1^* = u(x, y) - iv(x, y)$ are both analytic, prove that both $u(x, y)$ and $v(x, y)$ are constants.

Section-III

Marks=30

1. Show that:

$$\int_0^{2\pi} \frac{\sin^2 \theta}{a + b \cos \theta} d\theta = \frac{2\pi}{b^2} (a - \sqrt{a^2 - b^2}), \quad (a > b > 0).$$

2. Use the Cauchy's residue theorem to evaluate

$$\int_{|z|=3} \frac{e^z}{z^3 + 2z^2} dz.$$

3. Find the spherical polar coordinates components of the velocity of a moving particle. ($x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$)

4. Show that, in general coordinates, the quantities $\frac{\partial y^i}{\partial x^j}$ do not form the components of a tensor.

5. Show that (e_i) and (e^i) are reciprocal system of vectors. $\Rightarrow$ Page (8) PU-notes

PAPER: Mathematical Methods of Physics
Course Code: PHY-302

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.
SUBJECTIVE TYPE

Marks=20

Section-II (Short Questions)

1. If v_i are the components of a first-order tensor, show that $\nabla \cdot \mathbf{v} = \frac{\partial v_i}{\partial x_i}$ is a zero-order tensor.

2. Expand

$$f(z) = \frac{z - \sin z}{z^5}$$

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in a Laurent series valid for $0 < |z|$.

3. Use the divergence theorem to establish Green's identity

$$\iint_S (f \nabla g) \cdot \mathbf{n} dS = \iiint_V (f \nabla^2 g + \nabla f \cdot \nabla g) dV,$$

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where f and g are scalar functions with continuous second partial derivatives.

4. Show that covariant derivative of a covariant vector V_i is given by

$$V_{i;j} = \frac{\partial V_i}{\partial x^j} - V_k \Gamma_{ij}^k, \text{ where Christoffel symbol } \Gamma_{ij}^k = \epsilon^k_{ij} \frac{\partial}{\partial x^i} \epsilon_j^{k}.$$

5. Show whether or not the function $f(z) = \text{Re}(z)$ is analytic.

Section-III

Marks=30

1. Evaluate the Cauchy principal value of

$$\int_{-\infty}^{+\infty} \frac{x \sin x dx}{x^2 + 9} \quad (a > 0).$$

2. Use the Cauchy's residue theorem to evaluate

$$\oint_{|z-1|=2} \frac{\tan z}{z} dz.$$

3. Find the circular cylindrical components of the velocity and acceleration of a moving particle.

2-4-8 (At. Khan)

With $\hat{e}_1$ a unit vector in the direction of increasing θ_1 , show that

$$\nabla \cdot \hat{e}_1 = \frac{1}{h_1 h_2 h_3} \frac{\partial}{\partial q_1} (h_2 h_3).$$

$$\nabla \times \hat{e}_1 = \frac{1}{h_1} \left[\frac{\partial_2}{h_3} \frac{\partial}{\partial q_3} (h_1) - \frac{\partial_3}{h_2} \frac{\partial}{\partial q_2} (h_1) \right].$$

4. If $T_{i_1 i_2 \dots i_n}$ is a tensor of rank n , show that $\frac{\partial T_{i_1 i_2 \dots i_n}}{\partial x^{i_{n+1}}}$ is a tensor of rank $n+1$ (Cartesian coordinates).

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UNIVERSITY OF THE PUNJAB

Fifth Semester 2013
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Mathematical Methods of Physics-
Course Code: PHY-302

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Section-II (Short Questions)

6.4.2
1.4.1

1. Show that

$$\frac{1}{2\pi i} \oint_C z^{m-n-1} dz$$
 (with the contour encircling the origin once counterclockwise) is a representation of the Kronecker δ_{mn} .

$\frac{d}{dz} \left\{ \frac{z^m}{z^{m+1}} \right\}^2$ Marks=20
 $2^{m-m-1} = 2^0 = 1$
 $2^m \cdot 2^{-(m+1)} = 2^{-1} = \frac{1}{2}$

2. Expand

$$f(z) = \frac{z}{(z+1)(z-2)}$$

in a Laurent series valid for the annular domain $0 < |z-2| < 3$.

Q. 20. Ex. 19.3. Pg# 835, CH# 19
 $= \frac{1}{2\pi i} + \frac{0}{z-2}$

3. Use the divergence theorem to establish Green's identity

$$\iint_S (f \nabla g - g \nabla f) \cdot n dS = \iiint_V (f \nabla^2 g - g \nabla^2 f) dV$$

here f and g are scalar functions with continuous second partial derivatives.

4. If A is irrotational (i.e. $\nabla \times A = 0$), show that $A \times r$ is solenoidal (i.e. $\nabla \cdot (A \times r) = 0$). 1.8.3 (Arfken)

5. Verify that
 $\epsilon_{ijkl} = \epsilon_{jilk}$
 is an isotropic fourth-rank tensor.

4.2.4 (a)

Section-III

Marks=30

1. Evaluate the Cauchy principal value of

$$\int_{-\infty}^{+\infty} \frac{dx}{(x^2 + a^2)^2} \quad (a > 0).$$

2. Evaluate the trigonometric definite integral

$$\int_0^{2\pi} \frac{d\theta}{1 + b \cos \theta} \quad (|b| < 1)$$

CH. 19 } Pg# 843-849
 $z = a \tan \theta$
 CH. 19 } Pg# 963

3. Show that the quantities $g_{ij} = e_i \cdot e_j$ form the covariant components of a second-rank tensor. Pg# 968

4. Show that, in general coordinates, the quantities $\frac{\partial x^i}{\partial x^j}$ do not form the components of a tensor. Pg# 958

5. Calculate the elements g_{ij} of the metric tensor for spherical polar coordinates ($x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$). CH# 28 (TENSORS)